

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T08 · Fractions, Decimals and Percentages

With Answers

Questions with full worked solutions

■ 17 pages

FAMILY MAP

Fractions Decimals Percentages

F01 Convert a recurring decimal to a fraction

5 Q | 11 marks

F02 Convert terminating representations

2 Q | 4 marks

Working Unit 22

4MA1-2H Q18

ORIGINAL QUESTION

2 marks

18 Use algebra to show that $0.\dot{7}0\dot{2} = \frac{26}{37}$

(Total for Question 18 is 2 marks)

Worked Solution 22

Convert a recurring decimal to a fraction

UNIT 1
2 marks**Align a complete repeating block**

$$x = 0.\overline{702}, \quad 1000x = 702.\overline{702}.$$

Subtract and simplify

$$999x = 702 \Rightarrow x = \frac{702}{999} = \frac{26}{37}.$$

Final answer

$$\frac{26}{37}$$

Question 26

4MA1-2H Q15 (a)

ORIGINAL QUESTION UNIT 1

2 marks

15 (a) Use algebra to show that $0.3\dot{7}\dot{2} = \frac{41}{110}$

(2)

Worked solution 26

Convert a recurring decimal to a fraction

MODULAR UNIT 1

2 marks

(a) – Convert a recurring decimal to a fraction

Set up the shift

$$x = 0.372727\dots, \quad 100x = 37.272727\dots$$

Subtract and simplify

$$99x = 36.9 = \frac{369}{10} \Rightarrow x = \frac{369}{990} = \frac{41}{110}.$$

Final answer

41/110

Original question 24

4MA1-2H Q15 M01

MODULAR UNIT 1

2 marks

15 Use algebra to show that $0.\dot{7}\dot{6}\dot{3} = \frac{42}{55}$

Worked solution 24

Convert a recurring decimal to a fraction

MODULAR UNIT 1

2 marks

M01 – Convert a recurring decimal to a fraction**Shift and subtract**Let $x=0.763$ recurring; then $1000x-10x=763.63\dots-7.63\dots=756$.**Solve**

$$990x = 756 \Rightarrow x = 756/990 = 42/55.$$

Final answer

$$0.\overline{763} = \frac{42}{55}$$

16 Use algebra to show that the recurring decimal $0.28\dot{1}\dot{3} = \frac{557}{1980}$

Worked solution

June 2021 | Question 16 | 2 marks

Family: Convert a recurring decimal to a fraction • Variant: Non-recurring prefix followed by a recurring block

Method / working

1. Let $x=0.281313\dots$ and subtract shifted recurrences, e.g. $10000x-100x=2785$.

2. Solve $x=2785/9900=557/1980$.

Final answer: $557/1980$

- 18** $0.4\dot{x}$ is a recurring decimal.
 x is a whole number such that $1 \leq x \leq 9$

Find, in terms of x , the recurring decimal $0.4\dot{x}$ as a fraction.
Give your fraction in its simplest form.
Show clear algebraic working.

Worked solution

November 2021 | Question 18 | 3 marks

Family: Convert a recurring decimal to a fraction | Variant: Non-recurring prefix followed by a recurring block

METHOD / WORKING

1. Let
2. $y = 0.4\dot{x}$
3. So
4. $y = 0.4xxx...$
5. Multiply by 10 and 100:
6. $10y = 4.xxx...$
7. $100y = 4x.xxx...$
8. Subtract:
9. $100y - 10y = (4x.xxx...) - (4.xxx...)$
10. $90y = x + 36$
11. Therefore
12. $y = \frac{x + 36}{90}$
13. Answer: $\frac{x + 36}{90}$

FINAL ANSWER

$$\frac{x + 36}{90}$$

13 Use algebra to show that $0.\dot{6}\dot{8}\dot{1} = \frac{15}{22}$

(Total for Question 13 is 2 marks)

Answer reference | May-Nov 2020 | Question 13

Family: Convert terminating representations • Variant: Decimal to fraction or percentage

13	e.g. $x = 0.6\dot{8}\dot{1}$ and $100x = 68.\dot{1}\dot{8}$ or $10x = 6.\dot{8}\dot{1}$ and $1000x = 681.\dot{8}\dot{1}$			M1	e.g. two decimals that when subtracted give a finite decimal (must show understanding of recurring figures by 'dot' or at least 2 lots of 18 or 81 after the decimal point). Algebra required, use of any letter.
	$99x = 67.5, x = \frac{67.5}{99} = \frac{15}{22}$ or $990x = 675, x = \frac{675}{990} = \frac{15}{22}$ oe	show	2	A1	dep for completing the 'show that' arriving at given answer from correct working.
Total 2 marks					

Worked solution

May-Nov 2020 | Question 13 | 2 marks

Family: Convert terminating representations • Variant: Decimal to fraction or percentage

Official final mark-scheme pages are embedded immediately after this question.

16 Use algebra to show that $0.4\dot{3}\dot{8} = \frac{217}{495}$

(Total for Question 16 is 2 marks)

Answer reference | January 2023 | Question 16

Family: Convert terminating representations • Variant: Decimal to fraction or percentage

Question	Working	Answer	Mark	Notes
16	eg $1000x = 438.38\ldots$ $10x = 4.38\ldots$ or $100x = 43.838\ldots$ $x = 0.438\ldots$ oe		2	M1 For selecting 2 correct recurring decimals that when subtracted give a whole number or terminating decimal (43.4 or 434 etc) eg $1000x = 438.38\ldots$ and $10x = 4.38\ldots$ or $100x = 43.838\ldots$ and $x = 0.438\ldots$ with intention to subtract. (if recurring dots not shown then showing at least one of the numbers to at least 5sf) or $0.4 + 0.\dot{0}3\dot{8}$ and eg $1000x = 38.38\ldots$ & $10x = 0.3838\ldots$, with intention to subtract.
	eg $1000x - 10x = 438.38\ldots - 4.38\ldots = 434$ and $\frac{434}{990} = \frac{217}{495}$ or eg $100x - x = 43.838\ldots - 0.438\ldots = 43.4$ and $\frac{43.4}{99} = \frac{217}{495}$ or eg $1000x - 10x = 38.38\ldots - 0.3838 = 38$ and $0.4 + \frac{38}{990} = \frac{4 \times 99 + 38}{990} = \frac{434}{990} = \frac{217}{495}$ oe <i>working required</i>	Clearly shown		A1 For completion to $\frac{217}{495}$ dep on M1 and use of some algebra
Total 2 marks				

Worked solution

January 2023 | Question 16 | 2 marks

Family: Convert terminating representations • Variant: Decimal to fraction or percentage

Official final mark-scheme pages are embedded immediately after this question.